

Hadi Rayan Al-Sandid

Portfolio: hsandid.github.io

Github: github.com/hsandid

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EXPERIENCE

- **Cisco Systems** Paris, France
Software Engineer October 2023 - Current
 - Currently working on Cisco's SSE/SASE offerings (Secure Access, Secure Connect).
 - Contributor on the FD.io VPP (Vector Packet Processor) project
- **Cadence Design Systems** Sophia-Antipolis, France
Software Engineering Intern April 2023 - September 2023
 - Designed asynchronous logging system for Virtuoso with minimal performance impact, using C/C++/Bash
 - Adapted system to leverage relevant session information, resulting in up-to 200x increase in data collection.
- **Cisco Meraki** Paris, France
Software Engineering Intern April 2022 - August 2022
 - Implemented framework to enable local performance testing of Meraki data-planes, using C++/Python/Bash.
 - Conducted performance tests to estimate data-plane performance, and identify software bottlenecks and bugs.
- **American University of Beirut** Beirut, Lebanon
Research Assistant August 2019 - August 2021
 - Assisting a team of professors with the development of a Smart Irrigation System, which was a selected project in the Google AI Impact Challenge 2019.
 - This includes developing an on-site hardware solution to control irrigation routines, and also an Android mobile application to allow users to access agricultural field data.

EDUCATION

- **Ecole Polytechnique** Palaiseau, France
MSc&T, Internet-of-Things : Innovation and Management September 2021 - August 2023
GPA : 3.93/4.0
Courses: Computer Networking - Operating Systems - Distributed/Parallel Computing
- **American University of Beirut** Beirut, Lebanon
Bachelor of Engineering, Computer and Communications Engineering September 2017 - August 2021
GPA : 3.7/4.0, Graduated with Distinction
Courses: Software Architecture and Design - Embedded Systems - Cloud-based Services

PUBLICATION

- **Arrow: RISC-V Vector Accelerator for Machine Learning Inference**
Presented in the Fifth Workshop on Computer Architecture Research with RISC-V (CARRV 2021)
 - Participated to the development of Arrow, a design-time configurable hardware vector accelerator.
 - Experimented with the GNU/GCC RISC-V cross-compiler and the RISC-V v0.9 vector ISA extension to optimize code for Arrow
 - Developed benchmarks to test performance of ML-related instructions with Arrow, showing that it can execute subsets of vector and matrix benchmarks fundamental to machine learning inference 2x to 78x faster than a scalar RISC processor.

SKILLS SUMMARY

- **Programming:** Proficient in C/Go. Familiar with Python and Bash.
- **Networking:** CCNA Certified. Familiar with network configuration and routing on Linux.
- **Languages:** French (Native tongue), English (Professional proficiency), Arabic (Conversational)

INTERESTS AND PRESENTATIONS

- Presented at FOSDEM 2024 & co-authored Cisco Blog post, to advertise open-source contributions
- Involved in team sports and outdoor activities, including hiking, running, and football.
- Hobby projects in Game Development, and 3D Modeling/Animation